

The Digital Bretton Woods

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K E Y W O R D S

VAT Gap
Fiscal Sovereignty
Digital Public Infrastructure
OSS / IOSS
Deemed Supplier
Algorithmic Accountability
Preventive Enforcement
Permissioned Blockchain
Customs Interoperability
ViDA
Market Fairness

A B S T R A C T

The growth of cross-border digital commerce has exposed a structural weakness in how governments enforce VAT in platform-driven markets. Despite EU measures like the One-Stop Shop (OSS), the Import One-Stop Shop (IOSS), and the upcoming "VAT in the Digital Age" (ViDA) reforms, enforcement remains largely reactive-dependent on post-transaction audits and fragmented records across platforms, logistics, and customs. This paper argues that the VAT Gap is fundamentally a verifiability problem: platform algorithms increasingly make tax decisions without leaving tamper evident records that authorities can reliably verify. To address this, it proposes a federated permissioned blockchain framework that attaches a minimal "VAT Evidence Capsule" to each consignment. By linking VAT liability data to logistics identifiers in a privacy-preserving manner, this system enables real-time verification of VAT compliance at borders and other control points without requiring disclosure of proprietary platform data, shifting enforcement from reactive reconciliation to preventive verification. As part of a broader digital public infrastructure agenda, this model supports fiscal sovereignty, market fairness, and efficient cross-border trade. Aligned with the EU's ViDA roadmap, it provides a scalable path to reduce VAT leakage and strengthen trust in the digital economy.

1. Introduction

We stand at a defining moment in the history of the global economy. The digital revolution, which promised to democratize access to markets and information, has instead consolidated power within a handful of transnational platforms, creating a new economic order that can best be described as "techno feudalism". In this new reality, the traditional instruments of the sovereign state most critically, the capacity to levy and collect taxes for the public good are being eroded by a systemic inability to track, verify, and tax cross-border digital trade effectively. The "VAT Gap" in the European Union and globally is not merely an administrative inefficiency or a leakage of revenue; it is a structural failure of the current global economic architecture that undermines the social contract, distorts market competition, and starves our societies of the resources urgently needed to finance the Sustainable Development Goals (SDGs) and the green transition.

This report argues that the solution to the crisis of fiscal sovereignty is not simply "more enforcement" within the existing paradigm of reactive audits, but a fundamental architectural shift toward preventive, cryptographic verification. Drawing on the technical proposal for a "Federated Permissioned Blockchain" to enforce Value Added Tax (VAT) compliance, we outline the path toward a "Digital Bretton Woods"-a new framework for global economic governance that reasserts the primacy of public law over private

algorithmic fiat.

The analysis proceeds in six parts. First, we diagnose the crisis of the "analogue state" in a digital century, quantifying the VAT gap and its downstream effects on public services. Second, we deconstruct the mechanics of "techno feudalism" and the "visibility gap" that renders current enforcement mechanisms like the One-Stop Shop (OSS) and Import One-Stop Shop (IOSS) insufficient. Third, we present the technical solution: a federated ledger that anchors a "VAT Evidence Capsule" to every physical consignment at the moment of dispatch, creating an immutable, privacy-preserving record of tax liability. Fourth, we explore the concept of "Algorithmic Accountability," arguing that platforms must be held responsible for the fiscal outcomes of their routing and pricing logic. Fifth, we connect this reform to the broader agenda of financing the SDGs and establishing Digital Public Infrastructure (DPI) that serves the common good. Finally, we provide a concrete roadmap for implementation, aligned with the EU's "VAT in the Digital Age" (ViDA) package, to move from pilot to global standard.

The choice before us is stark: we can accept a future where the digital economy operates as an offshore haven of untaxed extraction, or we can build the infrastructure necessary to ensure that the wealth of the 21st century contributes to the civilization that creates it.

Part I: The Crisis of Fiscal Sovereignty in the 21st Century

1.1 The Erosion of the Westphalian Tax Base

For three centuries, the Westphalian system of nation-states relied on a simple premise: economic activity took place within physical borders, and the sovereign controlling those borders had the right and the ability to tax that activity. A factory, a store, or a warehouse had a physical nexus; it could be inspected, audited, and regulated. The tax base was tangible.

The digital revolution has dissolved this nexus. In the hyper-globalized platform economy, value chains are fragmented across multiple jurisdictions. A product may be designed in California, manufactured in Shenzhen, marketed on a platform hosted in Ireland, sold to a consumer in Berlin, and drop-shipped from a bonded warehouse in Liège. At each step, the transaction is dematerialized, turning into data flows that are easily manipulated or obscured.

The result is a "visibility gap". Tax authorities, designed for the analogue era, can observe the *outcomes* of trade-the declared tax returns but they cannot see the *causal chain* of the transaction. They lack real-time access to the data that determines tax liability; who the true seller of record is, what the true value of the good is, and where the consumption actually takes place.

This erosion of fiscal sovereignty is not an accident; it is a feature of the current digital architecture. As noted by economic observers, the ability of governments to tax the rich and powerful whether individuals or corporations has been hollowed out by a combination of lobbying, regulatory arbitrage, and technological opacity. The "VAT Gap" is the fiscal expression of this political failure.

1.2 Quantifying the Haemorrhage: The VAT Gap

The scale of this failure is staggering. The European Commission's own estimates place the VAT compliance gap-the difference between expected VAT revenue and the amount actually collected-at approximately EUR 89 billion to EUR 128 billion annually.

To put this figure in perspective:

EUR 128 billion is roughly equivalent to the entire annual budget of the European Union.

It exceeds the estimated annual investment gap for achieving the EU's 2030 climate targets in the energy sector.

It represents a massive transfer of wealth from the public purse to non-compliant actors, effectively subsidizing tax evasion.

The distribution of this gap is highly uneven. While some Member States like Finland and Austria maintain high compliance rates (gaps of 1-3%), others face gaps exceeding 20-30%. This divergence creates deep inequities within the Single Market, as compliant businesses in high-enforcement jurisdictions are forced to compete with goods flowing through "porous" borders in other Member States.

1.3 The Downstream Effects on Sustainable Development

The consequences of this leakage extend far beyond the balance sheet. Fiscal capacity is the lifeblood of the Sustainable Development Goals (SDGs). We are currently "alarmingly off-track" on the SDGs, with less than 20% of targets projected to be achieved by 2030. Achieving these goals requires massive public investment in healthcare, education, renewable energy, and digital infrastructure.

Yet, as we have argued elsewhere, the budget resources available for such multilateral and national efforts are a "pittance". While EU member states tax significant portions of their national income, the inability to capture revenue from the cross-border digital economy constrains their ability to invest in the future. The VAT gap represents funds that should be building schools, financing green hydrogen plants, and supporting the most vulnerable. Instead, these funds are lost to the "black box" of the platform economy.

Furthermore, this leakage exacerbates inequality. When the state cannot tax capital and digital trade effectively, it is forced to rely more heavily on taxing labour and consumption of immobile goods. This regressive shift places the burden of financing the state on workers and local SMEs, while the digital giants and cross-border arbitrageurs operate in a low-tax parallel reality. This fuels the populist backlash against globalization, as citizens rightly perceive that the system is rigged against them.

Part II: The Architecture of Techno feudalism and the Failure of Reactive Enforcement

2.1 Techno feudalism: The New Economic Order

To understand why current VAT enforcement fails, we must understand the structure of the market it seeks to regulate. We have moved beyond industrial capitalism into an era of "techno feudalism". In this system, digital platforms (marketplaces, search engines, social media) do not merely facilitate markets; they *are* the markets. They are the new feudal lords, controlling the digital land upon which all commerce must take place.

These platforms extract "rent" from every transaction-not just in monetary fees, but in data. They control the algorithms that determine which products are seen, which sellers succeed, and how prices are set. Crucially, they also control the *information* regarding tax liability. They know the true identity of the seller, the origin of the goods, and the real transaction price. However, under the current regulatory framework, they act as "gatekeepers" who monetize this access while deflecting the full responsibility for fiscal compliance.

The "Policy Paradox" described in recent technical analysis captures this tension: The EU legislates that VAT must be paid in the Member State of consumption, but the enforcement of this law depends on data that is created, transformed, and stored inside the proprietary, opaque systems of private platforms. The regulator is in the position of a vassal asking the lord for a truthful accounting of the harvest, without the independent means to verify the numbers.

2.2 The Limitations of OSS and IOSS

Recognizing the challenge of e-commerce, the EU introduced the

One-Stop Shop (OSS) and Import One-Stop Shop (IOSS) reforms in July 2021. These were significant steps forward. They simplified registration, allowing sellers to register in one Member State for all EU sales, and introduced the concept of the "deemed supplier," shifting liability from the underlying seller to the platform for certain imports $\leq$ EUR 150.

However, these reforms remain rooted in a reactive enforcement paradigm. They rely on:

Self-Declaration: The platform or seller declares the VAT collected.

Post-Transaction Audits: Tax authorities periodically check records to verify accuracy.

Border Trust: Customs authorities generally trust the IOSS identification number provided on the shipping label.

This architecture suffers from critical structural flaws, or "failure modes," that sophisticated actors exploit with ease:

Table 1: Structural Failures of the Current IOSS Regime

Field	Description	Function
Consignment ID	Hash-linked to the physical shipping label (GS1/SSCC).	Binds the digital record to the physical parcel.
Timestamp & Node ID	Precise time of dispatch and identity of the warehouse/platform.	Establishes the "origin of truth" and accountability.
Declared Value (Hash)	Cryptographic commitment to the taxable amount.	Prevents undervaluation; the value cannot be changed later.
Liability Flag	Boolean flag: Platform Deemed Supplier (Y/N).	Clarifies who owes the tax (platform vs. seller).
IOSS Token	Encrypted reference to the IOSS number.	Validates access to the simplified import regime.
Destination & Rate	Member State code and applied VAT rate.	Ensures correct allocation of revenue.

The European Court of Auditors has explicitly noted that these measures are insufficient to prevent VAT import fraud at scale. The system is built on "records and returns"-digital paperwork that can be forged or manipulated-rather than on "verifiable evidence" anchored to the physical reality of the goods.

2.3 The Economics of Non-Compliance

The current system creates a perverse incentive structure. Because enforcement is reactive and selective, the "cost" of evasion is low. A seller in a third country can systematically undervalue goods or

misclassify them. If caught, they simply disappear and re-emerge under a new digital identity-a phenomenon known as "phoenixing."

This creates a "market for lemons" scenario. Compliant EU SMEs, which cannot hide from their local tax authority, bear the full 20%+ VAT cost. Non-compliant foreign sellers can undercut them by the margin of the tax. This effectively subsidizes foreign production at the expense of domestic industry. The "fairness lens" is not rhetorical; it is hard economics. When the expected VAT burden differs systematically between compliant and non-compliant suppliers, price competition ceases to be about productivity and becomes about regulatory arbitrage.

Part III: The Proposal – A Federated Permissioned Blockchain for Global Trade

3.1 The Paradigm Shift: From Reconciliation to Prevention

To close the VAT gap and restore sovereignty, we must shift the paradigm from reactive auditing to preventive verification. We need a system where compliance is not checked *after* the parcel has been delivered and the seller has vanished, but is verified *before* the goods enter the single market.

This requires a new technological architecture: the Federated Permissioned Blockchain.

This is not a proposal for a "crypto-anarchist" public chain like Bitcoin, nor is it a call for a centralized EU database (which would raise data sovereignty and security concerns). It is a proposal for a shared, immutable ledger governed by a consortium of public authorities and trusted private actors.

3.2 The Mechanism: The VAT Evidence Capsule

The core innovation is the "VAT Evidence Capsule." This is a digital twin of the physical consignment, created at the moment of dispatch. It anchors the fiscal attributes of the trade to the logistics chain.

According to the technical specifications , the capsule contains a minimal set of privacy-preserving data:

Table 2: The Schema of the VAT Evidence Capsule

Failure Mode	Mechanism	Consequence
Undervaluation	Sellers declare goods below the EUR 150 threshold to utilize IOSS and avoid customs duties.	Revenue loss; distortion of trade statistics; unfair competition.
Identity Hijacking	Fraudulent sellers obtain valid IOSS numbers of compliant platforms and use them to clear illicit goods.	Compliant platforms face liability for sales they didn't make; fiscal authorities lose revenue.
Data Fragmentation	The marketplace, payment processor, and logistics carrier each hold different data segments.	No "single version of the truth" exists for the auditor; reconciliation is costly and incomplete.
The Audit Gap	Audit capacity is linear (human auditors), while parcel volume is exponential.	Enforcement becomes probabilistic (<1% checked), making evasion a rational economic choice.

3.3 How It Works: The "Green Lane" Logic

Dispatch: When a consumer buys a product on a platform, the platform's system calculates the VAT. Instead of just storing this in a private database, it generates the Evidence Capsule and "publishes" it to the Federated Ledger.

Commitment: The ledger records the capsule. It is now immutable. The platform cannot change the declared value later without leaving a tamper-evident trail.

Border Crossing: When the parcel arrives at EU customs, the officer scans the barcode. Their system queries the Federated Ledger.

Verification: The ledger returns a binary signal: VERIFIED or INVALID.

Verified: The capsule exists, the IOSS number is valid and belongs to the platform that minted the capsule, and the declared value matches the customs declaration. The parcel is released instantly (Green Lane).

Invalid: No capsule exists, or the data mismatches (e.g., the label says EUR 10 but the ledger commitment implies EUR 50). The parcel is stopped for inspection.

This system moves enforcement from probabilistic (catching 1% of cheats) to deterministic (gating 100% of unverified flows). It solves the "visibility gap" by forcing the upstream data (from the platform) to be inextricably linked to the downstream movement (the logistics).

3.4 Why Blockchain? The Case for Distributed Trust

Why not use a standard API or a centralized database? The answer lies in the geopolitical and commercial reality of "Techno feudalism."

Trust: Tax authorities in the EU do not trust data from third-country platforms blindly. Platforms do not trust competitors or foreign governments with their raw commercial data. A centralized database creates a "honeypot" for hackers and a single point of failure.

Federation: A federated blockchain allows multiple sovereign entities (Member States, Partner Countries) and private actors (Platforms, Logistics) to maintain the ledger jointly. No single party owns the truth; they share the maintenance of the truth.

Interoperability: This aligns with the World Customs Organization (WCO) research on blockchain, which highlights its potential to improve compliance and trade facilitation by creating a "single version of the truth" across borders.

Furthermore, using Zero-Knowledge Proofs (ZKPs), the system can verify that "VAT was paid on the correct value" without revealing the exact wholesale price or the consumer's identity to every node in the network. This balances the state's need for verification with the commercial need for privacy and GDPR compliance.

Part IV: Algorithmic Accountability and the "Computed" Liability

4.1 The Black Box Problem

In the digital economy, tax liability is increasingly "computed." It is the output of an algorithm that determines: Is this seller a business or a hobbyist? Is the buyer a taxable person? Which VAT rate applies (standard, reduced, zero)? Who is the deemed supplier?

Currently, these algorithms operate inside the "black box" of the platform. Regulators cannot see the logic; they only see the result. If the algorithm is biased-intentionally or accidentally-to minimize tax (e.g., by misclassifying goods or routing them through low-tax regimes), the state loses revenue. This is a form of "algorithmic unaccountability".

4.2 Enforcing Accountability through the Ledger

The proposed VAT Evidence Capsule enforces accountability by creating an immutable audit trail of the algorithm's decisions. By requiring the platform to commit to the *inputs* (item category, value, destination) and the *outputs* (VAT amount, liability status) on the ledger, we create a record that can be audited against the actual law.

This aligns with the broader push for "Algorithmic Accountability Acts" and regulation of AI. Just as we demand that algorithms driving cars or diagnosing diseases be safe and explainable, we must demand that algorithms determining fiscal obligations be verifiable. The ledger provides the "proof of computation" necessary to hold the "feudal lords" of the digital economy responsible for their automated decisions.

4.3 The "Deemed Supplier" as a Digital Sheriff

The "Deemed Supplier" rule was a crucial legal innovation, concentrating liability on the platforms. The blockchain proposal operationalizes this legal concept. It turns the platform into a "Digital Sheriff" that must mint a digital certificate for every transaction. If a platform allows fraudulent sellers to trade without minting valid capsules, the platform's own reputation score on the network degrades, and its "Green Lane" privileges can be revoked.

This shifts the burden of policing the millions of micro-sellers from the tax authority to the platform-the only entity with the data and technical capacity to do so. It aligns economic incentives with compliance.

Part V: Toward a Digital Bretton Woods – Governance and Geopolitics

5.1 The Need for New Global Institutions

The challenges we describe-VAT leakage, techno feudalism, algorithmic opacity-are not unique to Europe. They are global. Yet, our institutions for economic governance are relics of the post-WWII era. We need a "Digital Bretton Woods".

Just as the original Bretton Woods conference in 1944 established the IMF and World Bank to manage the industrial global economy, a Digital Bretton Woods must establish the frameworks for the

digital economy. Key pillars of this new order must include:

Data Sovereignty: The principle that nations have a right to the data generated by their citizens and economic activity. The Federated Ledger operationalizes this by ensuring that tax data regarding French citizens is accessible to French authorities, regardless of where the platform is headquartered.

Digital Public Infrastructure (DPI): We must stop treating digital infrastructure as a private monopoly. Payment systems, identity frameworks, and tax verification ledgers should be Digital Public Infrastructure-open, interoperable standards built for the public interest. The UN and UNDP have championed DPI as a governance enabler; this proposal applies that concept to fiscal policy.

North-South Equity: Developing nations suffer most from tax base erosion. They lack the audit capacity of the EU. A standardized, blockchain-based verification system would provide a "compliance in a box" solution for the Global South, allowing African and Latin American nations to capture revenue from digital giants without needing to build bespoke, expensive systems.

5.2 Geopolitical Alignment

This development presents a strategic opportunity for the European Union to shape global norms in digital trade governance. By leading the creation of a Protocol for Value Added Tax Verification, the EU can extend the "Brussels Effect" into the fiscal domain, setting a standard that exporters worldwide must follow to access the Single Market. As with GSM and GDPR, the size of the EU market would naturally encourage global adoption without the need for formal international agreements.

For the United States, such a protocol would function less as regulation and more as a trade facilitation tool. A transparent verification system would allow US platforms and exporters to clear EU customs faster and with fewer inspections, aligning with broader goals of supply chain efficiency and digitized trade documentation.

For China, as a major source of low-value e-commerce consignments and a leader in blockchain-enabled trade initiatives, the system offers a way to differentiate legitimate exporters from fraudulent actors. A "Green Lane" for verifiable goods would ensure smoother access to EU markets for compliant Chinese sellers while discouraging misuse of e-commerce channels.

For the Global South, particularly G77 and African Union countries, the protocol offers a practical tool to combat tax base erosion without requiring advanced audit infrastructure. By adopting a shared verification standard, these countries could strengthen domestic resource mobilization and address illicit financial flows as part of their digital public infrastructure efforts.

In this way, VAT verification evolves from a regional enforcement measure into a foundation for broader, cooperative digital trade governance.

Part VI: The Roadmap – Implementation and the

"Readiness Index"

6.1 Alignment with "VAT in the Digital Age" (ViDA)

The European Commission's "VAT in the Digital Age" (ViDA) package is currently the legislative vehicle for reform. It includes pillars for Digital Reporting Requirements (DRR), the Platform Economy, and Single VAT Registration.

Timeline:

2025: Adoption of the package and immediate e-invoicing authorization for Member States.

2028: Extension of the "Deemed Supplier" regime and Single VAT Registration.

2030: Mandatory e-invoicing for cross-border B2B and platform economy requirements.

2035: Full rollout.

The Blockchain proposal presented here is complementary to ViDA. While ViDA focuses heavily on intra-EU B2B (e-invoicing), our proposal targets the B2C Import flow, which is the high-risk vector for fraud. The "VAT Evidence Capsule" is effectively an "e-invoice for the package." We urge the Commission and Council to integrate the *technical standard* of the federated ledger into the implementation acts of ViDA, particularly for the 2028 and 2030 milestones.

6.2 The VAT Verification Readiness Index (VVRI)

"You cannot manage what you cannot measure." To support a shift toward verification-based VAT enforcement, we propose a VAT Verification Readiness Index (VVRI) to assess how prepared the ecosystem-customs, platforms, and logistics operators-is for real-time verification. This index would measure:

Data Completeness: Percentage of inbound parcels carrying a valid VAT Evidence Capsule on the ledger, indicating integration with the verification system.

Consistency: Rate of mismatch between capsule data and physical customs inspection, reflecting data integrity and fraud risk.

Timeliness: Speed at which verification responses are received at the border, determining the feasibility of a "Green Lane."

Audit Efficiency: Revenue safeguarded or recovered per enforcement hour, showing the productivity gains of preventive verification over reactive audits.

The VVRI would function as a practical dashboard for EU leaders to benchmark Member States, identify operational gaps, and hold platforms accountable for their role in the verification ecosystem.

6.3 The Pilot Phase

We recommend an immediate pilot project during 2026–2027 to test the operational feasibility of the VAT verification model in a

controlled, high-impact environment.

Corridor: The pilot should focus on a high-volume e-commerce trade lane such as China → Belgium/Netherlands, where large volumes of low-value consignments already enter the EU through major logistics hubs. This ensures meaningful data, realistic stress conditions, and measurable outcomes within a short period.

Scope: The pilot should be limited to consignments valued at ≤ EUR 150 operating under the IOSS regime, as this segment represents the highest fraud risk and the largest volume of parcels. Keeping the scope narrow allows for focused testing without disrupting broader customs operations.

Partners: Participation should include two major global marketplaces, two global logistics integrators, and three Customs Administrations from entry-point Member States. This combination ensures that platform data, logistics handling, and customs verification can be tested end-to-end within the real supply chain.

Method – Dual Recording: During the pilot, trade flows continue under existing procedures. However, for each transaction, a “shadow” VAT Evidence Capsule is created and recorded on the federated ledger alongside the normal customs declaration. No enforcement action is taken based on the capsule during the test phase. Instead, authorities compare the “ledger truth” (what the platform committed at dispatch) with the “customs declaration truth” (what was declared at entry). This allows precise measurement of undervaluation, misclassification, and identity misuse without disrupting trade. At the same time, customs can measure how quickly verified parcels could be cleared under a simulated “Green Lane,” demonstrating potential time savings and efficiency gains.

This pilot structure allows policymakers to generate empirical evidence of VAT leakage, validate the technical feasibility of real-time verification, and quantify both revenue protection and trade facilitation benefits before wider rollout.

Part VII: The Economic Dividend – Financing Our Future

7.1 Closing the Gap

The economic case is overwhelming. Closing the EUR 128 billion VAT gap is not just about balancing books; it is about funding civilization.

EUR 100 billion/year could finance the entire "Just Transition Mechanism" to support regions moving away from coal.

It could double the budget for Erasmus+ and Horizon Europe, driving innovation and youth mobility.

It provides the "Own Resources" the EU desperately needs to repay the Next Generation EU debt without increasing member state contributions.

7.2 Restoring Market Fairness

Beyond the revenue gains, the reform restores market fairness at a structural level. European SMEs and compliant businesses currently operate at a systematic disadvantage because they cannot escape the visibility of their domestic tax authorities. They price their goods with full VAT included, while non-compliant sellers operating through cross-border platforms can undercut them by the exact margin of the avoided tax.

This distortion is not a matter of efficiency or competitiveness but of unequal enforcement. By making VAT liability verifiable for every consignment before it enters the market, the reform removes this asymmetry. Price competition can once again be driven by innovation, product quality, service, and productivity rather than the ability to exploit regulatory gaps. Restoring this balance is essential to maintaining confidence in the fairness and legitimacy of the Single Market, particularly among small and medium enterprises that form its economic backbone.

7.3 A Moral Imperative

The case for reform also extends beyond economics into the realm of social responsibility. In a global context defined by widening inequality, climate pressures, and geopolitical instability, the erosion of the tax base directly weakens the capacity of states to serve their citizens. When digital platforms and cross-border actors are able to sidestep fiscal obligations that ordinary businesses and individuals must bear, the social contract is undermined. Public trust in institutions declines when citizens perceive that rules apply unevenly.

Tax evasion in this context is not abstract or victimless. Its consequences are visible in underfunded healthcare systems, strained education budgets, limited social protection, and insufficient investment in sustainable development. The resources lost through VAT leakage are resources that could finance green transitions, support vulnerable populations, and invest in future generations. Ensuring fair and verifiable VAT compliance is therefore not only an administrative reform but a step toward reinforcing social justice, shared responsibility, and long-term collective well-being.

Conclusion: The Call to Action

The technology to solve the VAT gap exists. The Federated Permissioned Blockchain is not science fiction; it is a practical application of existing tools to a pressing public problem. It represents a shift from the "Analog State" to the "Smart State"-a state that uses technology to enforce its laws and protect its citizens.

Commit to the "Digital Bretton Woods" agenda.

Invest in the Digital Public Infrastructure of the Federated Ledger.

Implement the VAT Evidence Capsule standard within the ViDA framework.

Let us move from the "visibility gap" to the "verifiable truth." Let us reclaim our fiscal sovereignty. Let us build an economy that works for everyone, not just the digital few.

The time for action is now.

Detailed Technical Addendum: The Mechanics of the Federated Ledger

To satisfy the technical scrutiny of this audience, we provide a deeper dive into the architectural decisions underpinning this proposal.

A. Consensus Mechanism: Proof of Authority (PoA)

A Proof of Authority model is particularly suited to this application because participants in the network are known, regulated institutions rather than anonymous actors. Unlike energy-intensive Proof of Work systems or token-based Proof of Stake mechanisms, PoA derives trust from legal authority and institutional accountability.

Validators: Only authorized entities such as Tax Authorities, Customs Administrations, and certified technical providers are permitted to validate blocks. This ensures that governance of the ledger remains within a trusted public-private framework.

Identity: Each validator is identifiable and legally accountable for its participation, aligning the network's operation with public law and auditability requirements.

Performance: PoA enables very high transaction throughput and near-instant finality, which is essential in logistics environments where verification decisions must occur within seconds at the border without delaying cargo flow.

B. Privacy Preservation: Zero-Knowledge Proofs (ZKPs)

Protecting commercially sensitive data-such as pricing strategies, supplier relationships, and transaction details is a critical requirement for adoption by global platforms.

The Problem: Writing raw transaction data to a shared ledger could expose sensitive commercial information to other participants on the network, including competitors or logistics partners.

The Solution: Zero-Knowledge Proofs allow a platform to mathematically prove that VAT was correctly calculated and declared without revealing the underlying value or business data. The ledger records proof of compliance rather than the confidential details themselves, ensuring verification while preserving privacy and adhering to GDPR principles.

C. Interoperability with the WCO Data Model

The system is designed to map directly to the World Customs Organization (WCO) Data Model (v3.11+). The data fields in the VAT Evidence Capsule correspond to existing customs elements such as customs value, goods shipment identifiers, and declaration references. This ensures that the blockchain enhances existing customs risk management systems rather than creating a parallel administrative layer, allowing authorities to integrate verification into current workflows with minimal disruption.

D. The Role of Smart Contracts

Smart contracts embedded in the ledger automate the legal logic behind VAT liability, particularly the "Deemed Supplier" rule

applied under IOSS.

Code Logic: For example, *If (Value ≤ 150 AND Destination Country = FR) → Apply IOSS logic at 20% VAT rate.*

Execution: This rule executes automatically at the moment the Evidence Capsule is created.

Result: VAT liability is assigned deterministically at the source, reducing ambiguity and simplifying subsequent audits or inspections.

E. Scalability and Energy Consumption

Because the system operates on a permissioned, non-mining architecture, its energy consumption is minimal-comparable to that of a standard cloud database. This avoids the environmental drawbacks associated with first-generation blockchains while still delivering the benefits of immutability, distributed trust, and resilience.

This report is submitted as a contribution to the global debate on economic reform, reflecting the urgent need to align our fiscal institutions with the technological realities of the 21st century.

References

1. Technofeudalism - The Libby App by Overdrive: Free ebooks & audiobooks from the library, <https://share.libbyapp.com/title/10429707>
2. Technofeudalism - Penguin Random House Retail, <https://www.penguinrandomhouseretail.com/book/?isbn=9781685891244>
3. Financing International Cooperation - Jeffrey D. Sachs, <https://www.jeffsachs.org/newspaper-articles/ze3d5b3yre96kfr4ayxtt5lkjns>
4. A Decade After Their Adoption at the UN, the World Remains Highly Committed to the Sustainable Development Goals, <https://sdgtransformationcenter.org/static/news/press-release-sustainable-development-report-2025/english-pr.pdf>
5. Market concentration implications of foundation models: - Gov AI, https://cdn.governance.ai/Market_Concentration_Implications_of_Foundation_Models.pdf
6. A/80/179-S/2025/468 General Assembly Security Council - the United Nations, <https://docs.un.org/en/A/80/179>
7. VAT in the Digital Age (ViDA) Adopted, <https://marosavat.com/vat-news/vat-in-the-digital-age>
8. U.S. Governed By Campaign Contributions, Can't Have A Good Fiscal Policy!: Economist Jeffrey Sachs - YouTube, <https://www.youtube.com/watch?v=KCEbUcfZopY>
9. VAT Gap - Taxation and Customs Union - European Commission https://taxation-customs.ec.europa.eu/taxation/vat/fight-against-vat-fraud/vat-gap_en
10. European Commission releases report on tax gaps to support competitiveness and fairer tax systems - Taxation and Customs Union, https://taxation-customs.ec.europa.eu/news/european-commission-releases-report-tax-gaps-support-competitiveness-and-fairer-tax-systems-2025-12-11_en
11. The EU's Questionable VAT Policy - Tax Foundation, <https://taxfoundation.org/blog/eu-vat-compliance-gap/>

12. Jeffrey Sachs - Wikiquote, https://en.wikiquote.org/wiki/Jeffrey_Sachs

13. Unveiling the Potential of Blockchain for Customs, https://www.wcoomd.org/-/media/wco/public/global/pdf/topics/research/research-paper-series/45_votaro_okazaki_unveiling_the_potential_of_blockchain_for_customs.pdf

14. WORLD CUSTOMS ORGANIZATION ORGANISATION MONDIALE DES DOUANES Digital Customs Outline of a concept paper on the use of blockchain, <https://www.wcoomd.org/-/media/wco/public/global/pdf/topics/facilitation/ressources/permanent-technical-committee/241-242/pc0732e.pdf?la=en>

15. Sharing the Algorithm: The Tax Solution to Generative AI - Scholarly Commons | George Washington University Law School Research, https://scholarship.law.gwu.edu/cgi/viewcontent.cgi?article=3077&context=faculty_publications

16. Algorithms (Part II) - After the Digital Tornado, <https://www.cambridge.org/core/books/after-the-digital-tornado/algorithms/3A5E44187B6C94E2954C2C1B031937AD>

17. THE DIGITAL ECONOMY AND SANCTIONS: HOW GEOPOLITICAL PRESSURE IS FUELING ALTERNATIVE ECONOMIC ECOSYSTEMS - FGV Europe, https://www.fgv-europe.de/wp-content/uploads/2025/12/relatorio_the_digital_economy_and_sanctions_ap1-1.pdf

18. From political slavery to social friendship - The Pontifical Academy of Social Sciences, https://www.pass.va/en/publications/studia-selecta/studia_selecta_06_pass/beliz.html
/improving-data-sharing-with-blockchain/

19. The Global Digital Governance Map - New America, <https://www.newamerica.org/planetary-politics/reports/governing-the-digital-future/the-global-digital-governance-map/>

20. The Africa Roundtable N°2 - Global Perspectives Initiative, <https://globalperspectives.org/en/the-africa-roundtable-n2/>

21. The State of Broadband 2022: Accelerating broadband for new realities - ITU, https://www.itu.int/dms_pub/itu-s/opb/pol/S-POL-BROADBAND.26-2022-PDF-E.pdf

22. INDUSTRIAL DEVELOPMENT REPORT 2026 - UNIDO, <https://www.unido.org/sites/default/files/unido-publications/2025-11/UNIDO%20IDR26.pdf>

23. Sachs op-ed: Financing health and education for all - the United Nations, <https://www.un.org/sustainabledevelopment/blog/2016/05/sachs-op-ed-financing-health-and-education-for-all/>

24. EU proposal for VAT in the Digital Age package formally adopted - KPMG Meijburg, <https://www.meijburg.com/news/eu-proposal-vat-digital-age-package-formally-adopted>

25. VAT in the Digital Age (ViDA) - Taxation and Customs Union, https://taxation-customs.ec.europa.eu/taxation/vat/vat-digital-age-vida_en

26. VAT in the Digital Age | Deloitte Netherlands, <https://www.deloitte.com/nl/en/services/tax/perspectives/vat-in-the-digital-age.html>

27. Quotes by Jeffrey D. Sachs (Author of The End of Poverty) - Goodreads, https://www.goodreads.com/author/quotes/6440.Jeffrey_D_Sachs

28. Improving data sharing with blockchain - WCO - World Customs Organization, <https://mag.wcoomd.org/magazine/wco-news-94>